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Introduction to Linear State Space Models

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Discrete-Time Linear State Space Models

For linear, time-invariant systems, a discrete-time *state-space model* looks like a *vector first-order finite-difference* model:

$$\begin{aligned} \underline{x}(n+1) &= \mathbf{A}\,\underline{x}(n) + \mathbf{B}\,\underline{u}(n) \\ \underline{y}(n) &= \mathbf{C}\,\underline{x}(n) + \mathbf{D}\,\underline{u}(n) \end{aligned}$$

where

- $\underline{x}(n) \in \mathbf{R}^N$ = *state vector* at time n
- $\underline{u}(n) = p \times 1$ vector of inputs
- $\underline{y}(n) = q \times 1$ output vector
- $\mathbf{A} = N \times N$ *state transition matrix*
- $\mathbf{B} = N \times p$ *input coefficient matrix*
- $\mathbf{C} = q \times N$ *output coefficient matrix*
- $\mathbf{D} = q \times p$ *direct path coefficient matrix*

The state-space representation is especially powerful for

- *multi-input, multi-output* (MIMO) linear systems
- *time-varying* linear systems
  (every matrix can have a time subscript n)

Zero-State Impulse Response (Markov Parameters)

Linear State-Space Model:

$$\begin{aligned} \underline{y}(n) &= \mathbf{C}\underline{x}(n) + \mathbf{D}\underline{u}(n) \\ \underline{x}(n+1) &= \mathbf{A}\underline{x}(n) + \mathbf{B}\underline{u}(n) \end{aligned}$$

The zero-state *impulse response* of a state-space model is easily found by direct calculation: Let $\underline{x}(0) \stackrel{\Delta}{=} \underline{0}$ and $\underline{u} = \mathbf{I}_p\delta(n) = \mathsf{diag}(\delta(n), \ldots, \delta(n))$. Then

$$\begin{aligned} \mathbf{h}(0) &= \mathbf{C}\underline{x}(0)\mathbf{B} + \mathbf{D}\,\mathbf{I}_p\delta(0) = \mathbf{D} \\ \underline{x}(1) &= \mathbf{A}\,\underline{x}(0) + \mathbf{B}\,\mathbf{I}_p\delta(0) = \mathbf{B} \\ \mathbf{h}(1) &= \mathbf{CB} \\ \underline{x}(2) &= \mathbf{A}\,\underline{x}(1) + \mathbf{B}\,\delta(1) = \mathbf{AB} \\ \mathbf{h}(2) &= \mathbf{CAB} \\ \underline{x}(3) &= \mathbf{A}\,\underline{x}(1) + \mathbf{B}\,\delta(1) = \mathbf{A}^2\mathbf{B} \\ \mathbf{h}(3) &= \mathbf{CA}^2\mathbf{B} \\ &\vdots \\ \mathbf{h}(n) &= \boxed{\mathbf{CA}^{n-1}\mathbf{B}}, \quad n > 0 \end{aligned}$$

Zero-State Impulse Response (Markov Parameters)

Thus, the "impulse response" of the state-space model can be summarized as

$$\boxed{\mathbf{h}(n) = \begin{cases} \mathbf{D}, & n = 0 \\ \mathbf{C}\mathbf{A}^{n-1}\mathbf{B}, & n > 0 \end{cases}}$$

- Initial state $\underline{x}(0)$ assumed to be $\underline{0}$
- Input "impulse" is $\underline{u} = \mathbf{I}_p \delta(n) = \text{diag}(\delta(n), \ldots, \delta(n))$
- Each "impulse-response sample" $\mathbf{h}(n)$ is a $p \times q$ matrix, in general
- The impulse-response terms $\mathbf{C}\mathbf{A}^n\mathbf{B}$ for $n \geq 0$ are called *Markov parameters*

Linear State-Space Model Transfer Function

- Recall the linear state-space model:

$$\begin{aligned} \underline{y}(n) &= \mathbf{C}\,\underline{x}(n) + \mathbf{D}\,\underline{u}(n) \\ \underline{x}(n+1) &= \mathbf{A}\,\underline{x}(n) + \mathbf{B}\,\underline{u}(n) \end{aligned}$$

  and its "impulse response"

$$\mathbf{h}(n) = \begin{cases} \mathbf{D}, & n = 0 \\ \mathbf{C}\mathbf{A}^{n-1}\mathbf{B}, & n > 0 \end{cases}$$

- The *transfer function* is the z transform of the impulse response:

$$\mathbf{H}(z) \stackrel{\Delta}{=} \sum_{n=0}^{\infty} \mathbf{h}(n) z^{-n} = \mathbf{D} + \sum_{n=1}^{\infty} \left(\mathbf{C}\mathbf{A}^{n-1}\mathbf{B}\right) z^{-n}$$

$$= \mathbf{D} + z^{-1}\mathbf{C} \left[\sum_{n=0}^{\infty} \left(z^{-1}\mathbf{A}\right)^n\right] \mathbf{B}$$

  The closed-form sum of a matrix geometric series gives

$$\boxed{\mathbf{H}(z) = \mathbf{D} + \mathbf{C}\left(z\mathbf{I} - \mathbf{A}\right)^{-1}\mathbf{B}}$$

  (a $p \times q$ *matrix* of rational polynomials in z)

- If there are p inputs and q outputs, then $\mathbf{H}(z)$ is a $p \times q$ *transfer-function matrix* (or "matrix transfer function")
- Given transfer-function coefficients, many digital filter *realizations* are possible (different computing structures)

Example: ($p = 3$, $q = 2$)

$$\mathbf{H}(z) = \begin{bmatrix} z^{-1} & \frac{1 - z^{-1}}{1 - 0.5z^{-1}} & 1 + z^{-1} \\ \frac{2 + 3z^{-1}}{1 - 0.1z^{-1}} & \frac{1 + z^{-1}}{1 - z^{-1}} & \frac{(1 - z^{-1})^2}{(1 - 0.1z^{-1})(1 - 0.2z^{-1})} \end{bmatrix}$$

System Poles

Above, we found the transfer function to be

$$\mathbf{H}(z) = \mathbf{D} + \mathbf{C}\left(z\mathbf{I} - \mathbf{A}\right)^{-1}\mathbf{B}$$

The poles of $\mathbf{H}(z)$ are the same as those of

$$H_p(z) \stackrel{\Delta}{=} \left(z\mathbf{I} - \mathbf{A}\right)^{-1}$$

By *Cramer's rule* for matrix inversion, the denominator polynomial for $\left(z\mathbf{I} - \mathbf{A}\right)^{-1}$ is given by the *determinant*:

$$d(z) \stackrel{\Delta}{=} |z\mathbf{I} - \mathbf{A}|$$

where $|\mathbf{Q}|$ denotes the *determinant* of the square matrix $\mathbf{Q}$ (also written as $\det(\mathbf{Q})$.)

- In linear algebra, the polynomial $d(z) = |z\mathbf{I} - \mathbf{A}|$ is called the *characteristic polynomial* for the matrix $\mathbf{A}$
- The roots of the characteristic polynomial are called the *eigenvalues* of $\mathbf{A}$
- Thus, the *eigenvalues* of the state transition matrix $\mathbf{A}$ are the system *poles*
- Each *mode of vibration* gives rise to a pole pair

Initial-Condition Response

Going back to the time domain, we have the linear discrete-time state-space model

$$\begin{aligned} \underline{y}(n) &= \mathbf{C}\,\underline{x}(n) + \mathbf{D}\,\underline{u}(n) \\ \underline{x}(n+1) &= \mathbf{A}\,\underline{x}(n) + \mathbf{B}\,\underline{u}(n) \end{aligned}$$

and its "impulse response"

$$\mathbf{h}(n) = \begin{cases} \mathbf{D}, & n = 0 \\ \mathbf{C}\mathbf{A}^{n-1}\mathbf{B}, & n > 0 \end{cases}$$

Given zero inputs and initial state $\underline{x}(0) \neq \underline{0}$, we get

$$\underline{y}_x(n) = \mathbf{C}\mathbf{A}^n\underline{x}(0), \quad n = 0, 1, 2, \ldots .$$

By *superposition* (for LTI systems), the *complete response* of a linear system is given by the sum of its *forced response* (such as the impulse response) and its *initial-condition response*

Difference Equation to State Space Form

A digital filter is often specified by its *difference equation* (Direct Form I). Second-order example:

$$y(n) = u(n) + 2u(n-1) + 3u(n-2) - \frac{1}{2}y(n-1) - \frac{1}{3}y(n-2)$$

Every nth order *difference equation* can be reformulated as a *first order vector* difference equation called the "state space" (or "state variable") representation:

$$\begin{aligned} \underline{x}(n+1) &= \mathbf{A}\,\underline{x}(n) + \mathbf{B}\,u(n) \\ y(n) &= \mathbf{C}\,\underline{x}(n) + \mathbf{D}\,u(n) \end{aligned}$$

For the above example, we have, as we'll show,

$$\mathbf{A} \stackrel{\Delta}{=} \begin{bmatrix} -\frac{1}{2} & -\frac{1}{3} \\ 1 & 0 \end{bmatrix} \quad \text{(state transition matrix)}$$

$$\mathbf{B} \stackrel{\Delta}{=} \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \text{(matrix routing input to state variables)}$$

$$\mathbf{C} \stackrel{\Delta}{=} \begin{bmatrix} 3/2 \\ 8/3 \end{bmatrix} \quad \text{(output linear-combination matrix)}$$

$$\mathbf{D} \stackrel{\Delta}{=} 1 \quad \text{(direct feedforward coefficient)}$$

Converting to State-Space Form by Hand

1. First, determine the filter transfer function $\mathbf{H}(z)$. In the example, the transfer function can be written, by inspection, as

$$\mathbf{H}(z) = \frac{1 + 2z^{-1} + 3z^{-2}}{1 + \frac{1}{2}z^{-1} + \frac{1}{3}z^{-2}}$$

2. If $\mathbf{h}(0) \neq 0$, we must "pull out" the parallel delay-free path:

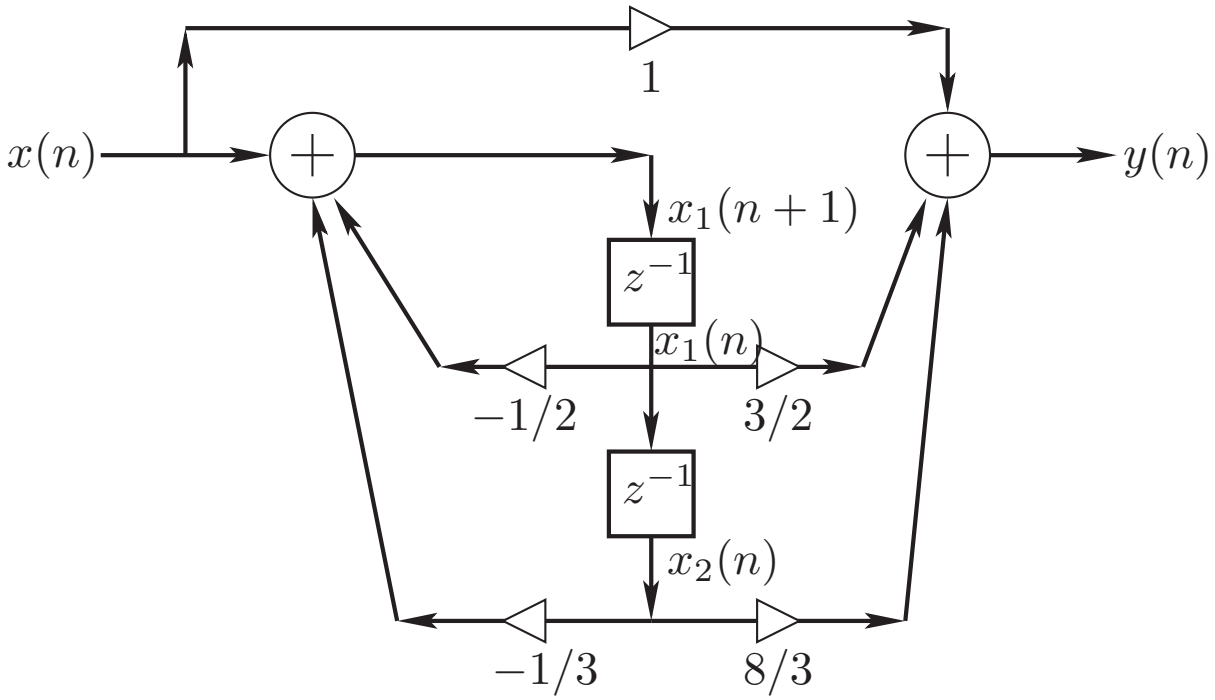
$$\mathbf{H}(z) = d_0 + \frac{b_1 z^{-1} + b_2 z^{-2}}{1 + \frac{1}{2}z^{-1} + \frac{1}{3}z^{-2}}$$

Obtaining a common denominator and equating numerator coefficients yields

$$\begin{aligned} d_0 &= 1 \\ b_1 &= 2 - \frac{1}{2} = \frac{3}{2} \\ b_2 &= 3 - \frac{1}{3} = \frac{8}{3} \end{aligned}$$

The same result is obtained using long or synthetic division

3. Next, draw the strictly causal part in *direct form II*, as shown below:



   It is important that the filter representation be *canonical with respect to delay*, i.e., the number of delay elements equals the order of the filter

4. Assign a state variable to the output of each delay element (see figure)

5. Write down the state-space representation by inspection. (Try it and compare to answer above.)

Matlab Conversion from Direct-Form to State-Space Form

Matlab has extensive support for state-space models, such as

- `tf2ss` - transfer-function to state-space conversion
- `ss2tf` - state-space to transfer-function conversion

Note that these utilities are documented primarily for continuous-time systems, but they are also used for discrete-time systems.

Let's repeat the previous example using Matlab:

Previous Example Using Matlab

```
>> num = [1 2 3]; % transfer function numerator
>> den = [1 1/2 1/3]; % denominator coefficients
>> [A,B,C,D] = tf2ss(num,den)

A =
   -0.5000   -0.3333
    1.0000         0

B =
     1
     0

C =  1.5000    2.6667

D =  1

>> [N,D] = ss2tf(A,B,C,D)

N = 1.0000    2.0000    3.0000

D = 1.0000    0.5000    0.3333
```

Matlab Documentation

The `tf2ss` and `ss2tf` functions are documented at
`http://www.mathworks.com/access/helpdesk/help/toolbox/signal/tf2ss.shtml`
as well as within Matlab itself (e.g., `help tf2ss`).

Related Signal Processing Toolbox functions include

- `tf2sos` — Convert digital filter transfer function parameters to second-order sections form.
- `sos2ss` — Convert second-order filter sections to state-space form.
- `tf2zp` — Convert transfer function filter parameters to zero-pole-gain form.
- `zp2ss` — Convert zero-pole-gain filter parameters to state-space form.

Similarity Transformations

A *similarity transformation* of a state-space system is a *linear change of state variable coordinates*:

$$\underline{x}(n) \stackrel{\Delta}{=} \mathbf{E}\underline{\tilde{x}}(n)$$

where

- $\underline{x}(n) =$ original state vector
- $\underline{\tilde{x}}(n) =$ state vector in *new coordinates*
- $\mathbf{E} =$ any *invertible* (one-to-one) matrix (linear transformation)

Substituting $\underline{x}(n) = \mathbf{E}\underline{\tilde{x}}(n)$ gives

$$\begin{aligned} \mathbf{E}\underline{\tilde{x}}(n+1) &= \mathbf{A}\,\mathbf{E}\underline{\tilde{x}}(n) + \mathbf{B}\underline{u}(n) \\ \underline{y}(n) &= \mathbf{C}\mathbf{E}\underline{\tilde{x}}(n) + \mathbf{D}\underline{u}(n) \end{aligned}$$

Premultiplying the first equation above by $\mathbf{E}^{-1}$ gives

$$\begin{aligned} \underline{\tilde{x}}(n+1) &= \left(\mathbf{E}^{-1}\mathbf{A}\mathbf{E}\right)\underline{\tilde{x}}(n) + \left(\mathbf{E}^{-1}\mathbf{B}\right)\underline{u}(n) \\ \underline{y}(n) &= (\mathbf{C}\mathbf{E})\,\underline{\tilde{x}}(n) + \mathbf{D}\underline{u}(n) \end{aligned}$$

Define the *transformed system matrices* by

$$\begin{aligned} \tilde{\mathbf{A}} &= \mathbf{E}^{-1}\mathbf{A}\mathbf{E} \\ \tilde{\mathbf{B}} &= \mathbf{E}^{-1}\mathbf{B} \\ \tilde{\mathbf{C}} &= \mathbf{C}\mathbf{E} \\ \tilde{\mathbf{D}} &= \mathbf{D} \end{aligned}$$

We can now write

$$\begin{aligned} \underline{\tilde{x}}(n+1) &= \tilde{\mathbf{A}}\underline{\tilde{x}}(n) + \tilde{\mathbf{B}}\underline{u}(n) \\ \underline{y}(n) &= \tilde{\mathbf{C}}\underline{\tilde{x}}(n) + \mathbf{D}\underline{u}(n) \end{aligned}$$

The transformed system describes the *same system* in new state-variable coordinates

Let's verify that the transfer function has not changed:

$$\begin{aligned} \tilde{\mathbf{H}}(z) &= \tilde{\mathbf{D}} + \tilde{\mathbf{C}}(z\mathbf{I} - \tilde{\mathbf{A}})^{-1}\tilde{\mathbf{B}} \\ &= \mathbf{D} + (\mathbf{CE})\left(z\mathbf{I} - \mathbf{E}^{-1}\mathbf{AE}\right)^{-1}(\mathbf{E}^{-1}\mathbf{B}) \\ &= \mathbf{D} + \mathbf{C}\left[\mathbf{E}\left(z\mathbf{I} - \mathbf{E}^{-1}\mathbf{AE}\right)\mathbf{E}^{-1}\right]^{-1}\mathbf{B} \\ &= \mathbf{D} + \mathbf{C}\left(z\mathbf{I} - \mathbf{A}\right)^{-1}\mathbf{B} = \mathbf{H}(z) \end{aligned}$$

- Since the eigenvalues of $\mathbf{A}$ are the poles of the system, it follows that the eigenvalues of $\tilde{\mathbf{A}} = \mathbf{E}^{-1}\mathbf{AE}$ are the same. In other words, eigenvalues are unaffected by a similarity transformation.
- The transformed Markov parameters, $\tilde{\mathbf{C}}\tilde{\mathbf{A}}^n\tilde{\mathbf{B}}$, are also unchanged since they are given by the inverse z transform of the transfer function $\tilde{\mathbf{H}}(z)$. However, it is also easy to show this by direct calculation.

State Space Modal Representation

Diagonal state transition matrix = *modal representation*:

$$\begin{bmatrix} x_1(n+1) \\ x_2(n+1) \\ \vdots \\ x_{N-1}(n+1) \\ x_N(n+1) \end{bmatrix} = \begin{bmatrix} \lambda_1 & 0 & 0 & \cdots & 0 \\ 0 & \lambda_2 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \lambda_{N-1} & 0 \\ 0 & 0 & 0 & 0 & \lambda_N \end{bmatrix} \begin{bmatrix} x_1(n) \\ x_2(n) \\ \vdots \\ x_{N-1}(n) \\ x_N(n) \end{bmatrix} + \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_{N-1} \\ b_N \end{bmatrix} u(n)$$

$$y(n) = \mathbf{C}\underline{x}(n) + \mathbf{D}u(n)$$

(always possible when there are no repeated poles)

The N complex modes are *decoupled*:

$$\begin{aligned} x_1(n+1) &= \lambda_1 x_1(n) + b_1 u(n) \\ x_2(n+1) &= \lambda_2 x_2(n) + b_2 u(n) \\ &\vdots \\ x_N(n+1) &= \lambda_N x_N(n) + b_N u(n) \\ y(n) &= c_1 x_1(n) + c_2 x_2(n) + \cdots + c_N x_N(n) + \mathbf{D}u(n) \end{aligned}$$

That is, the diagonal state-space system consists of N *parallel one-pole systems*:

$$\begin{aligned} \mathbf{H}(z) &= \mathbf{C}(z\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D} \\ &= \boxed{\mathbf{D} + \sum_{i=1}^{N} \frac{c_i b_i z^{-1}}{1 - \lambda_i z^{-1}}} \end{aligned}$$

Finding the (Diagonalized) Modal Representation

The ith *eigenvector* $\underline{e}_i$ of a matrix $\mathbf{A}$ has the defining property

$$\mathbf{A}\underline{e}_i = \lambda_i \underline{e}_i,$$

where λ_i is the associated *eigenvalue*. Thus, the eigenvector $\underline{e}_i$ is *invariant* under the linear transformation $\mathbf{A}$ to within a (generally complex) scale factor λ_i.

An $N \times N$ matrix $\mathbf{A}$ typically has N eigenvectors.[1] Let's make a similarity-transformation matrix $\mathbf{E}$ out of the N eigenvectors:

$$\mathbf{E} = \left[\begin{array}{cccc} \underline{e}_1 & \underline{e}_2 & \cdots & \underline{e}_N \end{array} \right]$$

Then we have

$$\mathbf{AE} = \left[\begin{array}{cccc} \lambda_1\underline{e}_1 & \lambda_2\underline{e}_2 & \cdots & \lambda_N\underline{e}_N \end{array} \right] \stackrel{\Delta}{=} \mathbf{E}\mathbf{\Lambda}$$

where $\mathbf{\Lambda} \stackrel{\Delta}{=} \mathrm{diag}(\underline{\lambda})$ is a diagonal matrix having $\underline{\lambda} \stackrel{\Delta}{=} \left[\begin{array}{cccc} \lambda_1 & \lambda_2 & \cdots & \lambda_N \end{array} \right]^T$ along its diagonal. Premultiplying by $\mathbf{E}^{-1}$ gives

$$\boxed{\mathbf{E}^{-1}\mathbf{AE} = \mathbf{\Lambda}}$$

Thus, $\mathbf{E} = \left[\begin{array}{cccc} \underline{e}_1 & \underline{e}_2 & \cdots & \underline{e}_N \end{array} \right]$ is a similarity transformation that *diagonalizes the system*.

[1] When there are repeated eigenvalues, there may be only one linearly independent eigenvector for the repeated group. We will not consider this case and refer the interested reader to a Web search on "generalized eigenvectors," e.g., `http://en.wikipedia.org/wiki/Generalized_eigenvector` .